MSML640 Project 1 Proposal

Basketball Shot Trajectory Tracker

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Individual Project

1 Team Composition

This is an **individual project**.

2 Application Description

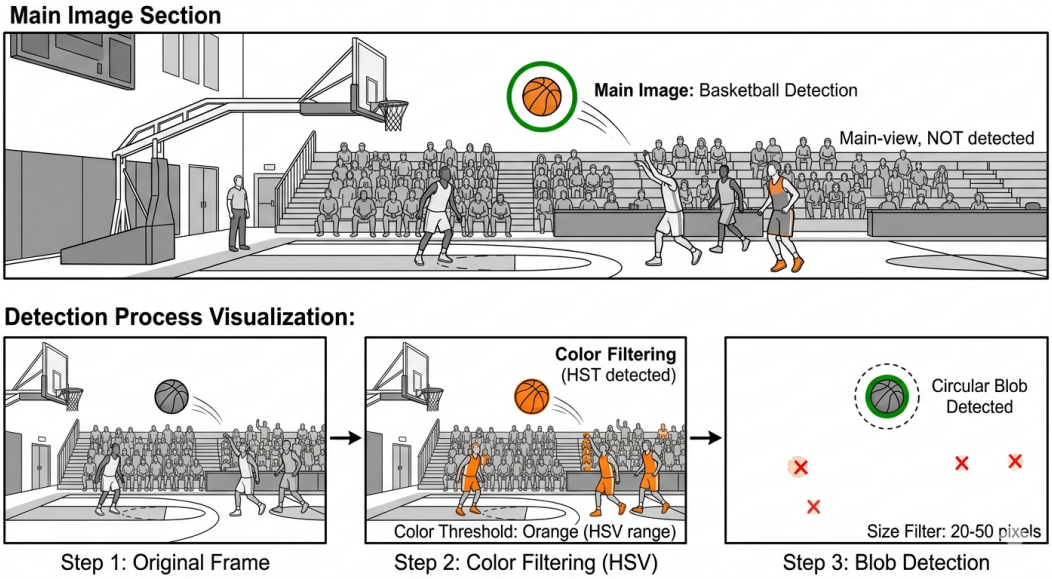

Figure 1: Conceptual diagram showing blob detection pipeline (generated by gemini)

This project develops a computer vision application that tracks basketball trajectories in real-world game footage, visualizing the flight path and calculating shot metrics (release angle, arc height). This explores how classical CV algorithms perform under challenging real-world conditions.

2.1 Core Algorithm

Blob detection identifies the basketball as a circular object in each frame. **Corner/interest point detection** maintains tracking when blob detection fails due to motion blur or occlusion. Tracked positions are connected across frames to reconstruct the ball's trajectory.

3 Dataset Description

3.1 Data Source

The dataset will consist of basketball game footage from multiple sources to ensure diversity:

- Publicly available YouTube clips (NBA, college, street ball)
- Self-captured footage from local basketball courts
- Creative Commons licensed sports footage (Pexels, Pixabay)
- 30-60 seconds per clip, multiple clips totaling 3-5 minutes
- Focus on side-angle camera views showing clear ball trajectory

3.2 Real-World Complexity

The footage contains authentic challenges: **motion blur** (fast-moving ball), **occlusion** (ball behind hoops/nets/players), **lighting variations** (indoor/outdoor), **camera movement** (panning/zooming), **cluttered backgrounds** (players, crowds, graphics), **false positives** (orange jerseys, shoes, markings), and **scale variation** (distance changes).

Dataset includes different shot types (free throws, three-pointers, layups), environments (arenas, gyms, outdoor courts), and lighting/camera qualities.

4 Expected Challenges & Technical Analysis

Preprocessing: Testing Gaussian blur levels, comparing HSV vs. RGB color spaces, implementing background subtraction, and developing frame normalization for lighting variations.

Parameter Tuning: Optimal blob size ranges, color threshold ranges for indoor/outdoor lighting, temporal smoothing window sizes, and confidence thresholds.

Robustness Testing: Ball spinning effects on feature tracking, tracking performance at different distances, recovery after tracking loss, and discrimination strategies for false positives.

Iterative Improvement: Five iterations documented: (1) baseline blob detection, (2) adaptive HSV thresholding, (3) size/motion filtering, (4) trajectory prediction during occlusion, (5) temporal smoothing. Each evaluated on tracking success rate with detailed failure analysis.

5 Significance

This project addresses practical sports analysis while providing rich technical depth for exploring classical CV algorithm behavior under real-world challenges, perfectly aligning with course objectives.